

Qualification Pack



Junior Cloud Computing Associate

QP Code: NIE/ITS/Q1202

Version: 1.0

NSQF Level: 4

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:h.sihare@nielit.gov.in

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NIE/ITS/Q1202: Junior Cloud Computing Associate

Brief Job Description

A Junior Cloud Computing Associate is an entry-level professional in the field of cloud computing. They assist in managing and maintaining cloud infrastructure and services. Their responsibilities may include basic cloud resource provisioning, monitoring, and support tasks.

Personal Attributes

It includes Virtualization, Basics of Cloud Computing, and Linux Operating System with the expertise to effectively implement virtualization technologies, grasp essential cloud computing concepts, and master the Linux operating system

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ITS/N1204: Conceptualize Operating Systems - Windows and Linux](#)
2. [NIE/ITS/N1205: Introduction to Network Management System](#)
3. [NIE/ITS/N1206: Implementing Virtualization and Basics of Cloud Computing Technology](#)
4. [NIE/ITS/N1207: Conceptualize Advance Linux](#)
5. [NIE/ITS/N1209: Emphasizing Cloud Computing Services](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
7. [NIE/ITS/N1208: Managing Cloud Architecture with Open Stack](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
Country	India
NSQF Level	4

Qualification Pack

Credits	15
Aligned to NCO/ISCO/ISIC Code	2511.9900
Minimum Educational Qualification & Experience	12th grade Pass with NA of experience OR Completed 2nd year of the 3-year diploma after 10 (and pursuing regular diploma) OR 10th grade pass (and pursuing continuous schooling) OR 10th Class (pass) with 2 Years of experience in relevant field OR Previous relevant Qualification of NSQF Level (Previous relevant Qualification of NSQF Level 3.0 with minimum education as 8th Grade pass) with 3 Years of experience in relevant field
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	29/03/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-04-IT-00354-2023-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/ITS/N1204: Conceptualize Operating Systems - Windows and Linux

Description

Operating systems are fundamental software components that manage and facilitate the interaction between hardware and software in a computer system. Windows and Linux are two prominent operating systems, each with its unique characteristics, architectures, and use cases

Scope

The scope covers the following :

- Able to operate operating Window & Linux Operating System

Elements and Performance Criteria

Operating System

To be competent, the user/individual on the job must be able to:

- PC1.** Configure Network Settings and perform user and group management
- PC2.** Knowledge of various management service roles and get acquainted with the concept of remote desktop management
- PC3.** Installation of Windows and Linux Operating System
- PC4.** Creation of disk partitions and configure Active directory
- PC5.**
 - Conceptualization of s of Group policy, work Shell and run various Windows
 - commands
- PC6.** Basic Commands in Linux and implement administrator tasks with sudo.
- PC7.** Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Window Operating System
- KU2.** Linux Operating System and its command

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand and navigate the Windows operating system, including its Control Panel and settings.
- GS2.** Install and configure Windows and Windows Server Operating Systems effectively.
- GS3.** Install and configure Windows and Windows Server Operating Systems effectively.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operating System</i>	25	17	5	-
PC1. Configure Network Settings and perform user and group management	-	-	-	-
PC2. Knowledge of various management service roles and get acquainted with the concept of remote desktop management	-	-	-	-
PC3. Installation of Windows and Linux Operating System	-	-	-	-
PC4. Creation of disk partitions and configure Active directory	-	-	-	-
PC5. • Conceptualization of s of Group policy, work Shell and run various Windows • commands	-	-	-	-
PC6. Basic Commands in Linux and implement administrator tasks with sudo.	-	-	-	-
PC7. Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk.	-	-	-	-
NOS Total	25	17	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1204
NOS Name	Conceptualize Operating Systems - Windows and Linux
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	2.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1204: Conceptualize Advance Linux.

Description

Advanced Linux refers to the utilization of Linux operating systems in a more sophisticated and specialized manner, often involving advanced concepts and techniques. It typically goes beyond basic usage and delves into areas like system administration, programming, and advanced system configurations. Here are some key elements and concepts associated with advanced Linux

Scope

The scope covers the following :

- Able to work on Linux & its commands

Elements and Performance Criteria

Linux Commands

To be competent, the user/individual on the job must be able to:

PC1. In-depth knowledge of advanced Utilities and Services in Linux

PC2. Setting up of different services of Linux like DNS Server, Web Server, Mail Server, File Server, and Squid Proxy Serve

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Setting Server using Linux OS

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Linux Commands</i>	5	25	-	-
PC1. In-depth knowledge of advanced Utilities and Services in Linux	-	-	-	-
PC2. Setting up of different services of Linux like DNS Server, Web Server, Mail Server, File Server, and Squid Proxy Serve	-	-	-	-
NOS Total	5	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1204
NOS Name	Conceptualize Advance Linux.
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud support services, Cloud Management
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	NA

Qualification Pack

NIE/ITS/N1205: Introduction to Network Management System

Description

Network Management is the practice of administering, monitoring, and maintaining a computer network to ensure its optimal performance, security, and reliability. It involves a set of activities, tools, and techniques used to control, plan, deploy, and optimize network resources, services, and infrastructure. Effective network management is crucial for organizations of all sizes to ensure their networks function efficiently and securely.

Scope

The scope covers the following :

- Able to configure and manage network system

Elements and Performance Criteria

Network Management

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks
- PC2.** Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topology used in networking
- PC3.** Explain various Transmission modes, Ethernet and the Classification Of Transmission Media.
- PC4.** In-depth idea about various networking devices like Repeaters, Hubs, Bridges,Switches and router
- PC5.** Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols
- PC6.** IP addressing, Classes, IP Addressing Scheme, Subnetting Basics along with How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM
- PC7.** Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Overview of TCP/IP & OSI Model
- KU2.** Network devices, IP Addressing & Topology configuration

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Understand and apply the OSI and TCP/IP models to manage network communication effectively.
- GS2.** Identify and configure various networking devices such as routers, switches, hubs, and bridges.
- GS3.** Explain and implement TCP/IP protocols, including understanding port numbers and common TCP/UDP ports.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Network Management</i>	25	17	5	-
PC1. Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks	-	-	-	-
PC2. Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topology used in networking	-	-	-	-
PC3. Explain various Transmission modes, Ethernet and the Classification Of Transmission Media.	-	-	-	-
PC4. In-depth idea about various networking devices like Repeaters, Hubs, Bridges,Switches and router	-	-	-	-
PC5. Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols	-	-	-	-
PC6. IP addressing, Classes, IP Addressing Scheme, Subnetting Basics along with How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM	-	-	-	-
PC7. Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing	-	-	-	-
NOS Total	25	17	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1205
NOS Name	Introduction to Network Management System
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud support services, Cloud Management
NSQF Level	4
Credits	2.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1205: Managing Cloud Architecture with Open Stack.

Description

OpenStack is an open-source cloud computing platform that allows you to build and manage your own private or public cloud infrastructure. It is designed to provide a flexible and scalable solution for managing cloud resources and services. Here's an overview of how you can manage cloud architecture with OpenStack

Scope

The scope covers the following :

- Configure and manage openstack Services

Elements and Performance Criteria

Openstack

To be competent, the user/individual on the job must be able to:

- PC1.** Basic components and Architectures of open Stack along with Installation of Openstack
- PC2.** Configure and manage Open Stack Client and Horizon dashboard.
- PC3.** Configure and manage OpenStack Identity Management
- PC4.** Configure and manage Image management
- PC5.** Configure and manage Compute Service

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** OpenStack is an open-source cloud platform for managing public, private, and hybrid cloud infrastructures
- KU2.** Key OpenStack components include Nova (compute), Swift (object storage), Cinder (block storage), Neutron (networking), and Horizon (dashboard).
- KU3.** Keystone handles identity management, providing secure authentication, authorization, and role-based access control across OpenStack.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Effectively manage and configure cloud resources using OpenStack components like Nova, Swift, and Neutron.
- GS2.** Troubleshoot and resolve issues related to cloud infrastructure, ensuring optimal performance and reliability.
- GS3.** Implement security best practices, including identity management, role-based access control, and secure API usage.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Openstack</i>	5	25	-	-
PC1. Basic components and Architectures of open Stack along with Installation of Openstack	-	-	-	-
PC2. Configure and manage Open Stack Client and Horizon dashboard.	-	-	-	-
PC3. Configure and manage OpenStack Identity Management	-	-	-	-
PC4. Configure and manage Image management	-	-	-	-
PC5. Configure and manage Compute Service	-	-	-	-
NOS Total	5	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1205
NOS Name	Managing Cloud Architecture with Open Stack.
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	NA
Next Review Date	NA

Qualification Pack

NIE/ITS/N1206: Implementing Virtualization and Basics of Cloud Computing Technology

Description

Virtualization and Cloud Computing are two closely related concepts in modern IT infrastructure. They offer numerous benefits, including resource optimization, cost reduction, scalability, and flexibility. Here's an overview of each concept and how they are implemented:

Scope

The scope covers the following :

- Able to understand Cloud computing and its Virtualization methodology

Elements and Performance Criteria

Virtualization

To be competent, the user/individual on the job must be able to:

- PC1.** Differentiating between Open source Virtualization and proprietary Virtualization.
- PC2.** Installation and use Open source virtualization tools - KVM, Hyper-V and Xen
- PC3.** Creation of Virtual Machine using Virtualization Tools
- PC4.** Cloud Computing Concepts and would be able to differentiate Cloud computing vs Cluster computing vs. Grid computing.
- PC5.** Conceptualize Cloud Characteristics and Working of Cloud Computing along with features Of Cloud Computing
- PC6.** Knowledge pertaining to Types of Cloud
- PC7.** Enlightened about the Benefits of Cloud Computing and Applications of Cloud Computing In industries
- PC8.** In-depth coverage of Cloud Platform ranging from its benefits to its implementation and its scope and its pros and cons

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Virtualization and its type
- KU2.** Open source Virtualization and proprietary Virtualization.
- KU3.** Creation of Virtual Machines
- KU4.** Types of Cloud & its service model

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Understand and apply concepts of virtualization, including open-source and proprietary tools.
- GS2.** Install and manage virtualization tools like KVM, Hyper-V, and Xen for efficient resource use.
- GS3.** Configure and deploy virtual machines using virtualization platforms effectively.
- GS4.** Differentiate between cloud computing, cluster computing, and grid computing.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Virtulization</i>	50	32	10	-
PC1. Differentiating between Open source Virtualization and proprietary Virtualization.	-	-	-	-
PC2. Installation and use Open source virtualization tools - KVM, Hyper-V and Xen	-	-	-	-
PC3. Creation of Virtual Machine using Virtualization Tools	-	-	-	-
PC4. Cloud Computing Concepts and would be able to differentiate Cloud computing vs Cluster computing vs. Grid computing.	-	-	-	-
PC5. Conceptualize Cloud Characteristics and Working of Cloud Computing along with features Of Cloud Computing	-	-	-	-
PC6. Knowledge pertaining to Types of Cloud	-	-	-	-
PC7. Enlightened about the Benefits of Cloud Computing and Applications of Cloud Computing In industries	-	-	-	-
PC8. In-depth coverage of Cloud Platform ranging from its benefits to its implementation and its scope and its pros and cons	-	-	-	-
NOS Total	50	32	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1206
NOS Name	Implementing Virtualization and Basics of Cloud Computing Technology
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	4.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1206: Emphasizing Cloud Computing Services.

Description

Cloud computing services offer a wide range of benefits to businesses and individuals. Emphasizing these benefits can help in understanding the significance of cloud computing services.

Scope

The scope covers the following :

- Cloud Computing Service Model

Elements and Performance Criteria

Cloud Computing Service Model

To be competent, the user/individual on the job must be able to:

- PC1.** SaaS Cloud Deployment with Open Source Tools
- PC2.** Introduction to PaaS Cloud and SaaS Cloud Deployment with Google App Engine
- PC3.** Knowledge of IaaS Cloud, install and configure ESXi Server along with setting up of a Private Cloud With Vsphere

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** IaaS, PaaS & SaaS Model
- KU2.** VMware Virtualization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand and conceptualize Software as a Service (SaaS) for delivering cloud-based applications.
- GS2.** Conceptualize Platform as a Service (PaaS) for developing, deploying, and managing applications efficiently.
- GS3.** Understand Infrastructure as a Service (IaaS) for managing virtualized computing resources in the cloud.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cloud Computing Service Model</i>	5	25	-	-
PC1. SaaS Cloud Deployment with Open Source Tools	-	-	-	-
PC2. Introduction to PaaS Cloud and SaaS Cloud Deployment with Google App Engine	-	-	-	-
PC3. Knowledge of IaaS Cloud, install and configure ESXi Server along with setting up of a Private Cloud With Vsphere	-	-	-	-
NOS Total	5	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1206
NOS Name	Emphasizing Cloud Computing Services.
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	NA

Qualification Pack

NIE/ITS/N1207: Conceptualize Advance Linux

Description

Advanced Linux refers to the utilization of Linux operating systems in a more sophisticated and specialized manner, often involving advanced concepts and techniques. It typically goes beyond basic usage and delves into areas like system administration, programming, and advanced system configurations. Here are some key elements and concepts associated with advanced Linux

Scope

The scope covers the following :

- Able to work on Linux & its commands

Elements and Performance Criteria

Linux Commands

To be competent, the user/individual on the job must be able to:

PC1. In-depth knowledge of advanced Utilities and Services in Linux

PC2. Setting up of different services of Linux like DNS Server, Web Server, Mail Server, File Server, and Squid Proxy Serve

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. How to configure and manage DNS servers for domain name resolution in a network.

KU2. The process of setting up and configuring web servers for hosting websites and applications.

KU3. How to configure mail servers to manage email sending, receiving, and storage.

KU4. The steps to set up and manage file servers for efficient data storage and sharing.

KU5. How to configure Squid proxy servers to control internet access and improve network performance.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Configure and manage DNS servers to ensure efficient domain name resolution.

GS2. Set up and maintain web servers for hosting websites and web applications.

GS3. Manage mail servers for secure and reliable email communication.

GS4. Configure and administer file servers for effective data storage and sharing.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Linux Commands</i>	34	8	5	-
PC1. In-depth knowledge of advanced Utilities and Services in Linux	-	-	-	-
PC2. Setting up of different services of Linux like DNS Server, Web Server, Mail Server, File Server, and Squid Proxy Serve	-	-	-	-
NOS Total	34	8	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1207
NOS Name	Conceptualize Advance Linux
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud support services, Cloud Management
NSQF Level	4
Credits	1.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1207: Managing Cloud Architecture with Open Stack-

Description

OpenStack is an open-source cloud computing platform that allows you to build and manage your own private or public cloud infrastructure. It is designed to provide a flexible and scalable solution for managing cloud resources and services. Here's an overview of how you can manage cloud architecture with OpenStack

Scope

The scope covers the following :

- Configure and manage openstack Services

Elements and Performance Criteria

Openstack

To be competent, the user/individual on the job must be able to:

- PC1.** Basic components and Architectures of open Stack along with Installation of Openstack
- PC2.** Configure and manage Open Stack Client and Horizon dashboard.
- PC3.** Configure and manage OpenStack Identity Management
- PC4.** Configure and manage Image management
- PC5.** Configure and manage Compute Service

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** OpenStacks architecture, components, and cloud computing role
- KU2.** manage resources using OpenStack Client and Horizon dashboard.
- KU3.** OpenStack Identity Management for user authentication, authorization, and access control is essential.
- KU4.** configure image management and compute services for efficient cloud resource handling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply technical proficiency in cloud computing, OpenStack architecture, and virtualization.
- GS2.** Analyze and solve complex technical issues effectively using problem-solving strategies.
- GS3.** Communicate technical concepts clearly through documentation and team collaboration.
- GS4.** Work efficiently in teams, sharing knowledge and supporting peers in cloud projects.
- GS5.** Manage cloud infrastructure projects, from planning to execution, with efficiency.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Openstack</i>	34	8	-	-
PC1. Basic components and Architectures of open Stack along with Installation of Openstack	-	-	-	-
PC2. Configure and manage Open Stack Client and Horizon dashboard.	-	-	-	-
PC3. Configure and manage OpenStack Identity Management	-	-	-	-
PC4. Configure and manage Image management	-	-	-	-
PC5. Configure and manage Compute Service	-	-	-	-
NOS Total	34	8	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1207
NOS Name	Managing Cloud Architecture with Open Stack-
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1209: Emphasizing Cloud Computing Services

Description

Cloud computing services offer a wide range of benefits to businesses and individuals. Emphasizing these benefits can help in understanding the significance of cloud computing services.

Scope

The scope covers the following :

- Cloud Computing Service Model

Elements and Performance Criteria

Cloud Computing Service Model

To be competent, the user/individual on the job must be able to:

PC1. SaaS Cloud Deployment with Open Source Tools

PC2. Introduction to PaaS Cloud and SaaS Cloud Deployment with Google App Engine

PC3. Knowledge of IaaS Cloud, install and configure ESXi Server along with setting up of a Private Cloud With Vsphere

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. IaaS, PaaS & SaaS Model

KU2. VMware Virtualization

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Differentiate between SaaS, PaaS, and IaaS, understanding their roles in cloud environments.

GS2. Recognize key features like scalability, flexibility, and on-demand access in cloud computing.

GS3. Implement SaaS applications using open-source tools for efficient deployment.

GS4. Manage cloud-based applications to optimize performance and user experience.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cloud Computing Service Model</i>	33	8	-	-
PC1. SaaS Cloud Deployment with Open Source Tools	-	-	-	-
PC2. Introduction to PaaS Cloud and SaaS Cloud Deployment with Google App Engine	-	-	-	-
PC3. Knowledge of IaaS Cloud, install and configure ESXi Server along with setting up of a Private Cloud With Vsphere	-	-	-	-
NOS Total	33	8	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1209
NOS Name	Emphasizing Cloud Computing Services
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/ITS/N1208: Managing Cloud Architecture with Open Stack

Description

OpenStack is an open-source cloud computing platform that allows you to build and manage your own private or public cloud infrastructure. It is designed to provide a flexible and scalable solution for managing cloud resources and services. Here's an overview of how you can manage cloud architecture with OpenStack

Scope

The scope covers the following :

- Configure and manage openstack Services

Elements and Performance Criteria

Openstack

To be competent, the user/individual on the job must be able to:

- PC1.** Basic components and Architectures of open Stack along with Installation of Openstack
- PC2.** Configure and manage Open Stack Client and Horizon dashboard.
- PC3.** Configure and manage OpenStack Identity Management
- PC4.** Configure and manage Image management
- PC5.** Configure and manage Compute Service

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** OpenStacks architecture, components, and cloud computing role
- KU2.** manage resources using OpenStack Client and Horizon dashboard.
- KU3.** OpenStack Identity Management for user authentication, authorization, and access control is essential.
- KU4.** configure image management and compute services for efficient cloud resource handling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply technical proficiency in cloud computing, OpenStack architecture, and virtualization.
- GS2.** Analyze and solve complex technical issues effectively using problem-solving strategies.
- GS3.** Communicate technical concepts clearly through documentation and team collaboration.
- GS4.** Work efficiently in teams, sharing knowledge and supporting peers in cloud projects.
- GS5.** Manage cloud infrastructure projects, from planning to execution, with efficiency.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Openstack</i>	33	8	5	-
PC1. Basic components and Architectures of open Stack along with Installation of Openstack	-	-	-	-
PC2. Configure and manage Open Stack Client and Horizon dashboard.	-	-	-	-
PC3. Configure and manage OpenStack Identity Management	-	-	-	-
PC4. Configure and manage Image management	-	-	-	-
PC5. Configure and manage Compute Service	-	-	-	-
NOS Total	33	8	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1208
NOS Name	Managing Cloud Architecture with Open Stack
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1208: Managing Cloud Architecture with open stack - -

Description

OpenStack is an open-source cloud computing platform that allows you to build and manage your own private or public cloud infrastructure. It is designed to provide a flexible and scalable solution for managing cloud resources and services. Here's an overview of how you can manage cloud architecture with OpenStack

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Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Openstack</i>	33	8	5	-
PC1. Basic components and Architectures of open Stack along with Installation of Openstack	-	-	-	-
PC2. Configure and manage Open Stack Client and Horizon dashboard.	-	-	-	-
PC3. Configure and manage OpenStack Identity Management	-	-	-	-
PC4. Configure and manage Image management	-	-	-	-
PC5. Configure and manage Compute Service	-	-	-	-
NOS Total	33	8	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1208
NOS Name	Managing Cloud Architecture with open stack - -
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management, Cloud Management, Cloud support services
NSQF Level	4
Credits	1.5
Version	1.0
Next Review Date	NA

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

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7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1204.Conceptualize Operating Systems - Windows and Linux	25	17	5	0	47	13
NIE/ITS/N1205.Introduction to Network Management System	25	17	5	0	47	13
NIE/ITS/N1206.Implementing Virtualization and Basics of Cloud Computing Technology	50	32	10	0	92	27
NIE/ITS/N1207.Conceptualize Advance Linux	34	8	5	0	47	13
NIE/ITS/N1209.Emphasizing Cloud Computing Services	33	8	0	0	41	13
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	12
NIE/ITS/N1208.Managing Cloud Architecture with Open Stack	33	8	5	-	46	9
Total	220	120	30	-	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
On the Job Training	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf